

YOUSIF M. ELSHEIKH M.

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[website](#) / [linkedin](#) / [github](#) / [scholar](#)

PROFESSIONAL EXPERIENCE

- 09/2024 – Present Institute of Aircraft Systems, University of Stuttgart**
Research Associate / Stuttgart, Germany
Verification and validation of safety-critical cyber-physical systems and deployed neural networks.
- 12/2023 – 08/2024 IPG Automotive GmbH**
Simulation Backend Working Student / Karlsruhe, Germany
Developed SIL-based V2X communication in CarMaker. Implemented CPM messages and their sending, receiving, encoding, and decoding using abstract ASN.1.
- 11/2022 – 09/2023 Adam Riese GmbH**
Computer Vision Working Student / Stuttgart, Germany
Implemented YOLOv7 in a multi-stage object detection pipeline with ONNX Runtime. Trained, validated, benchmarked, and deployed the complete pipeline.
- 03/2022 – 09/2022 WIS (Würth Industrie Service GmbH & Co. KG)**
Machine Learning Development Intern / Bad Mergentheim, Germany
Developed 3D computer vision methods for object localization and recognition. Computed the Z dimension from point cloud data and confirmed the feasibility of direct point cloud learning with VoteNet.
- 07/2021 – 03/2022 Technical University of Kaiserslautern**
Student Assistant. Underwater object detection with FLS. Generated GPR sensor data using GANs and autoencoders.
- 08/2018 – 02/2019 Hi-Force Engineering Works**
Design Engineer. 3D CAD, standardization, and market research. Won a waterworks project valued at more than €100,000.
- 12/2016 – 12/2017 University of Khartoum**
Teaching Assistant. Tutorials in Fluid Mechanics I, Computer Applications, Measurement and Instrumentation and Hydraulic & Pneumatic Power Transmission.

EDUCATION

- 2024 Technical University of Kaiserslautern**
M.Sc. Mechanical and Process Engineering, specialization in Vehicle Engineering
Thesis: *Reinforcement Learning on Sensor Data and Reaction Grid Maps*, Robotics Research Lab.
- 2016 University of Khartoum**
B.Sc. Mechanical Engineering
Second Class, Division One.

SELECTED PUBLICATIONS

- SAFECOMP 2026 On the Timing Determinism of CUDA and Vulkan SC for Safety-critical AI Workloads on SoC Integrated GPUs**
Yousif M. Elsheikh*, Adrian Raiser*, Fabian Frommherz*, and Zamira Daw.
- IEEE IRAI 2026 Adversarial Sampling at Code Boundary**
Yassine Akhiat, Yousif M. Elsheikh, Henry Spaet, Zamira Daw, and Bjoern Annighoefer.

Skills

Core

Verification & Validation, Simulation, Reinforcement Learning, Computer Vision, Deep Learning, Autonomous Driving

Software

C/C++, Python, PyTorch, Stable Baselines, Pandas, NumPy, conda, Linux, Docker, Git, Car-Maker, Unreal Engine

CAD

SolidWorks, MATLAB + Simulink

Languages

Arabic Native
German C1 / DSH-2
English C1